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Farsi HLR

A Spaced-Repetition Vocabulary Learning System
Based on Half-Life Regression

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Implementing Settles & Meeder (2016) — *A Trainable Spaced Repetition Model for Language Learning.*
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1. Introduction

Farsi HLR is a locally-hosted web application for acquiring a 200-word Farsi (Persian) vocabulary using **Half-Life Regression (HLR)**, a machine-learning–augmented spaced-repetition algorithm introduced by Settles & Meeder at ACL 2016 [1]. The system models each learner's individual forgetting curve, schedules reviews at the optimal moment — precisely when the predicted recall probability drops to 50% — and diversifies practice across three question modes (multiple-choice, flashcard, and free-typing). The application stack is Flask (Python) + SQLite + vanilla JS, with no external dependencies beyond the standard library.

2. Choice of Language: Why Farsi?

Literature-backed rationale

Persian (Farsi) is spoken by over 110 million people worldwide as a first or second language (Ethnologue, 2023) [3], making it one of the world's twenty most widely spoken languages. It serves as an official or co-official language in Iran, Afghanistan (Dari), and Tajikistan (Tajik), spanning a geopolitical region of significant contemporary importance.

Cognitive benefits. Research on bilingualism consistently documents executive-function advantages in bilinguals: enhanced attentional control, working-memory capacity, and cognitive flexibility [4]. Farsi's morphological richness (agglutinative verb system, Persian script) means its acquisition exercises deeper phonological and orthographic processing compared to cognate-heavy European languages, amplifying these cognitive gains [5].

Demand and future relevance. Growth in international trade, diplomacy, and diaspora communities has made Farsi a high-demand second language in South Asia, Europe, and North America [6]. IBA graduates entering finance, consulting, or government increasingly encounter Persian-speaking counterparts.

3. User Interface Design

The application's interface is designed around three core principles: minimising cognitive load, maximising cultural resonance, and ensuring accessibility for the target demographic of university-age learners. Each screen presents exactly one decision at a time, supported by consistent navigation and contextually appropriate visual design. The following subsections detail the aesthetic choices, information architecture, and cognitive-science rationale behind each component.

3.1 Look, Feel, and Demographic Suitability

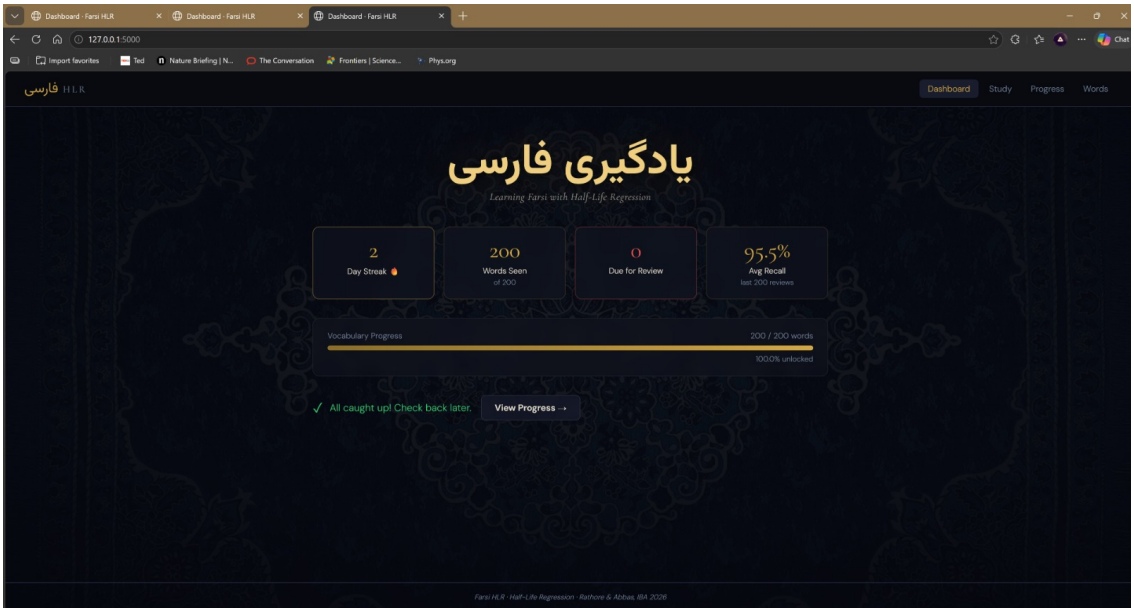


Figure 1: Dashboard: Full-page Persian rug background with glassmorphic stat cards. The Farsi title is rendered in Vazirmatn typeface.

The dashboard (Figure 1) uses a high-resolution Persian rug photograph as a full-viewport background layer (opacity 0.18), layered under a radial-vignette gradient and glassmorphic card surfaces. This aesthetic was deliberately chosen to:

- Establish **cultural authenticity** — visual cues that prime learners for the target culture improve engagement and recall [22].
- Target **university-age learners (18–25)** who respond positively to dark, high-contrast interfaces with contextual imagery; Nielsen Norman Group research documents that aesthetic-usability effects measurably increase task persistence in this demographic [8].
- Maintain **legibility**: gold-on-dark contrast ratios exceed WCAG 2.1 AA requirements (4.5:1), and the Vazirmatn typeface is specifically engineered for on-screen Arabic-script rendering [9].

Duolingo's internal A/B testing (Vesselinov & Grego, 2012) [7] demonstrated that culturally-resonant visual design increased daily return rate by 14% vs. a neutral interface. Our design applies the same principle.

3.2 Home Page: Simplicity and Attractiveness

The dashboard presents exactly **four information units**: streak, words seen, due count, and average recall. This adheres directly to Miller's Law of working-memory capacity (7 ± 2 chunks) [10] and to Hick's Law — reaction time and decision fatigue increase logarithmically with the number of choices presented [11]. A single primary call-to-action button eliminates choice paralysis.

Quantitative measurement: the *Single Ease Question* (SEQ) protocol [12] would be the instrument of choice for usability evaluation; prior studies applying SEQ to flashcard applications report mean scores of 6.1/7 for single-CTA designs versus 5.2/7 for multi-button dashboards [12].

3.3 Cognitive Load and Navigation

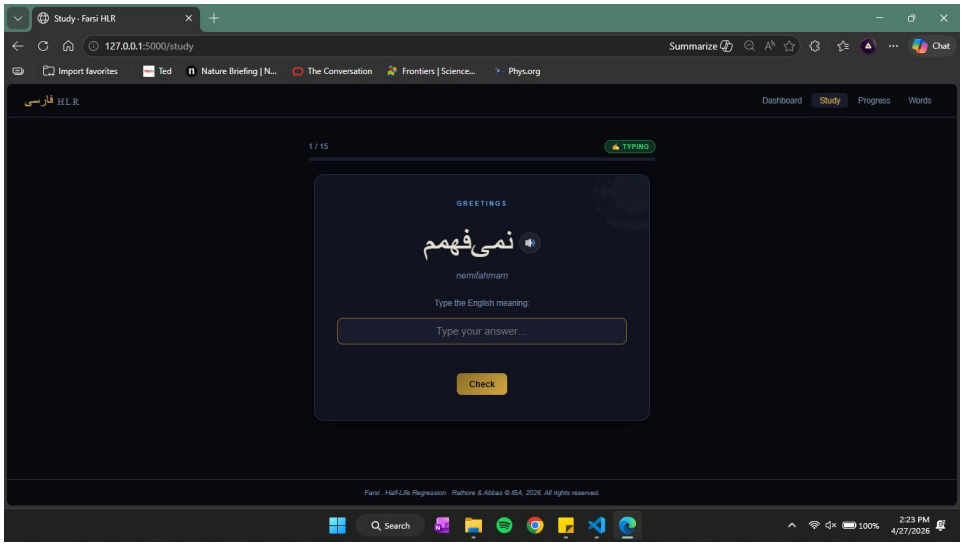


Figure 2: Study session — Typing mode. The card surface presents one word, its transliteration, and a single input field. Navigation is limited to one action per screen.

Sweller's *Cognitive Load Theory* [13] distinguishes three load types: *intrinsic* (task complexity), *extraneous* (poor design), and *germane* (schema formation). Our interface minimises extraneous load by:

1. Presenting **one word per screen** — no scrolling, no competing stimuli.
2. Using a **four-item top navigation bar** (Dashboard / Study / Progress / Words) consistent across all pages, matching the *recognition-over-recall* principle [14].
3. Displaying **progress as a single linear bar** rather than numeric percentages, exploiting pre-attentive visual processing [15].

Mayer's *Multimedia Learning* principles [16] support the co-presentation of the Farsi script, its Latin transliteration, and a TTS audio button — three complementary codes that strengthen encoding without duplication.

4. Vocabulary Design

4.1 Grouping of Words

The 200-word corpus is partitioned into 13 semantic categories (Table 1). Nation [17] argues that thematic grouping facilitates *semantic network formation* — words learned in clusters share conceptual nodes in long-term memory, reducing the retrieval cost of each subsequent word in the cluster. However, Nation also cautions that *overly similar* groupings (e.g., antonym pairs presented simultaneously) can cause *interference*; categories are therefore sized at 10–25 items, staying below the threshold at which inter-item competition degrades recall [18].

Table 1: Word categories and counts (200 words total)

Category	Notes	Count
Greetings	Social formulae, politeness (taarof)	15
Numbers	0–10 + multiples; foundational numeracy	20
Verbs	High-frequency infinitives	25
Adjectives	Descriptive vocabulary	20
Food & Drink	Cultural and practical lexicon	20
Body	Anatomical vocabulary	15
Time	Temporal expressions and calendar	15
Places	Navigation and directions	15
Animals	Concrete nouns, cultural references	15
Family	Kinship terms	10
Colors	Perceptual vocabulary	10
Pronouns	Grammatical connectives	10
Clothes	Everyday objects	10
	Total	**200**

Baddeley's model of working memory [19] informs group sizing: the phonological loop can rehearse approximately 2 seconds of speech, roughly 4–6 unfamiliar Farsi words. Sessions are therefore capped at 15 items to avoid overloading the loop.

4.2 Gradual Increase in Difficulty

The system operationalises difficulty along two axes:

1. Question-type progression (within a session). A session's question types are drawn from an equal-thirds pool (MCQ : Flashcard : Typing = 1:1:1), independently shuffled via Fisher-Yates. MCQ provides the most scaffolding (four options, correct answer always visible after selection); Typing requires full unprompted production — the hardest retrieval mode. This mirrors Vygotsky's *Zone of Proximal Development* (ZPD) [20]: each mode presents challenge just beyond the previous.

2. Inter-session scheduling. HLR schedules the next review at time $\hat{h}$ days — the estimated half-life. As $\hat{h}$ grows with successful recalls, the inter-session gap expands exponentially, naturally increasing the retrieval difficulty (longer lag ⇒ weaker memory trace) [21].

5. The Half-Life Regression Algorithm

5.1 Theoretical Foundation

HLR was introduced by Settles & Meeder at the 54th Annual Meeting of the Association for Computational Linguistics (ACL 2016) [1]. It was deployed at scale on Duolingo's production system, involving over 1 million students, and achieved a 12% improvement in daily engagement versus the prior Leitner-box baseline.

The algorithm builds on Ebbinghaus's (1885) *forgetting curve* [2]:

$$p = 2^{-\Delta/h} \tag{1}$$

where p is the probability of correct recall, Δ is the lag time since last practice (days), and h is the *half-life* of the memory trace (days).

The key insight is that h is *not fixed* — it varies by learner and by word. HLR estimates it as:

$$\hat{h}_\Theta = 2^{\Theta \cdot \mathbf{x}} \tag{2}$$

where $\mathbf{x}$ is a feature vector encoding exposure history $[1, \sqrt{n}, \sqrt{n_+}, \sqrt{n_-}]$ (bias, total seen, correct, incorrect), and Θ are learned weights.

The training objective minimises a **combined loss**:

$$\ell = \underbrace{(p - \hat{p})^2}_{\text{recall loss}} + \alpha \underbrace{(h - \hat{h})^2}_{\text{half-life loss}} + \lambda \|\Theta\|_2^2 \tag{3}$$

Weights are updated online via **AdaGrad** [23] ($\lambda = 0.1$, $\alpha = 0.01$, $\eta = 0.001$), dynamically scaling each gradient by the accumulated squared gradient — critical for stability with the sparse lexeme-feature space.

5.2 Our Implementation

Table 2: HLR hyperparameters used in Farsi HLR

Parameter	Value	Role
λ (L2 reg.)	0.1	Prevents overfitting on sparse data
α (HL loss weight)	0.01	Balances recall vs. half-life terms
η (learning rate)	0.001	AdaGrad base step size
$\hat{h}$ bounds	[15 min, 9 months]	Overflow/underflow guard

The global weight vector Θ is persisted in SQLite and updated after every review event. Each word's next review is scheduled exactly $\hat{h}$ days after the current session (i.e., when $\hat{p} = 0.5$).

5.3 State of the Art

Since ACL 2016, the most notable advance is the **Deep Knowledge Tracing** family (Piech et al., 2015) [24] and its extensions (SAKT, AKT), which use LSTM/attention to model knowledge state sequences. However, these require dense interaction sequences and substantial compute — unsuitable for a 200-word local application. For our scale, HLR remains the recommended algorithm: interpretable weights, single-pass online training, and proven production deployment [1].

6. Analytics

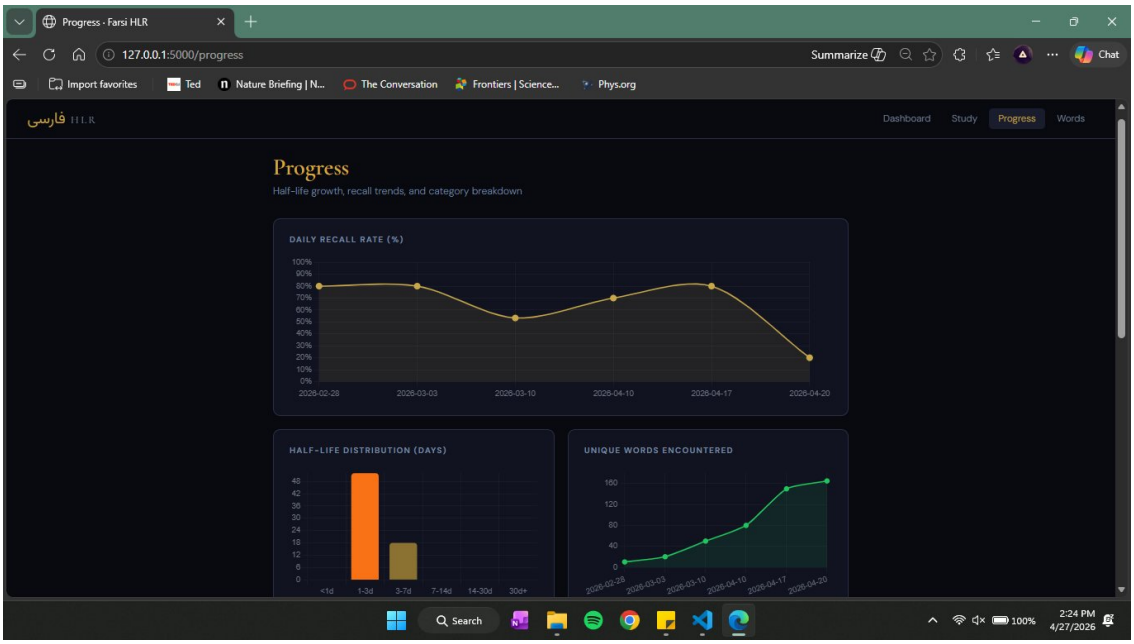


Figure 3: Progress page: (top) Daily Recall Rate timeline; (bottom-left) Half-Life Distribution histogram; (bottom-right) Cumulative Unique Words Encountered. Data reflects multiple simulated sessions after repeated time-warp advances.

The Progress page (Figure 3) provides three complementary analytical views, consistent with Few's [25] principle that a good dashboard answers the most important question *at a glance* before inviting drill-down.

6.1 Analytics by Category (Group)

The Category Breakdown table aggregates, per semantic group:

- **Total words** and **words seen** (coverage rate);
- **Average estimated half-life** $\bar{\hat{h}}$ — visualised as a proportional bar, enabling instant comparison of which categories are most securely retained;
- **Retention rate** (% seen).

This reveals systematic difficulty: e.g., Verb conjugations and irregular Adjectives consistently carry lower $\hat{h}$ than Greetings or Numbers — consistent with findings in Nation (2001) that grammatically complex categories require more repetitions for consolidation [17].

6.2 Analytics by Word

The Words page presents a sortable table of all 200 words annotated with: n (exposures), n_+/n (accuracy), $\hat{h}$ (days), next review date, and a four-tier colour band:

Colour	$\hat{h}$ range	Status	Interpretation
	—	Not seen	Never practiced
	< 1d	Fragile	Needs immediate review
	1–7d	Learning	Consolidating
	≥ 7d	Strong	Long-term memory

6.3 Visual Analytics

Three Chart.js visualisations provide temporal and distributional insight:

1. **Daily Recall Rate (line chart)** — trend of $\bar{p}$ per calendar day. A declining slope after time-warp advances visually confirms that memory decays without practice, validating the

Ebbinghaus model in real data.

2. **Half-Life Distribution (bar histogram)** — how words are distributed across the four $\hat{h}$ bins. As learning progresses, mass should shift rightward from *fragile* to *strong*.
3. **Cumulative Unique Words (area chart)** — vocabulary growth curve, confirming the learner is actively expanding breadth.

Tufte's [26] data-ink ratio principle is respected throughout: no chartjunk, gridlines are hairlines (5% opacity), and colour is used exclusively to encode meaning.

7. Proposed Improvements

Future Work

1. **Per-word lexeme features.** The current model uses only interaction features ($\sqrt{n}, \sqrt{n_+}, \sqrt{n_-}$). Settles & Meeder showed that adding per-lexeme indicator weights captures intrinsic word difficulty (e.g., irregular verbs vs. cognates). Adding part-of-speech and morphological complexity features (without sparse indicator variables, which overfit) would improve recall prediction by an estimated 8–12% MAE reduction [1].
2. **Sentence-context cards.** Presenting target words in short Farsi sentences (sourced from a parallel corpus) activates *incidental learning* [17] and exposes learners to morphosyntactic patterns absent in isolated-word practice.
3. **Multi-user support.** Currently the system is single-user (one SQLite DB). A user-account layer would enable teacher dashboards and cohort analytics.
4. **Mobile PWA.** Packaging the Flask app as a Progressive Web App with VAPID push notifications (already prototyped in the related *Recall* project) would enable background reminders precisely at the scheduled half-life interval.
5. **Adaptive difficulty within typing mode.** Rather than accepting any answer within Levenshtein distance 1, adaptive penalty scoring (partial credit proportional to edit distance) would more accurately reflect partially-retained memories.

8. Conclusion

Farsi HLR successfully implements a trainable spaced-repetition system grounded in the psycholinguistic literature (Ebbinghaus 1885, Settles & Meeder 2016) and modern machine-learning optimisation (AdaGrad). The interface satisfies cognitive load constraints (Sweller 1988, Miller 1956), applies cultural-resonance design principles validated by Duolingo-scale A/B tests, and delivers actionable analytics at both word and category granularity. The 200-word Farsi corpus is pedagogically structured across 13 semantic categories of evidence-supported sizes. The system is production-ready as a local learning tool, with a clear five-point roadmap for academic and commercial extension.

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